BALWYN HIGH SCHOOL

Student Name				
Teacher	GSL	HAL	LXI	SMH

Chemistry Unit 3

SAC 1: Fuels for Transport
Concepts from Area of Study 1

Friday 26th April, 2019

Reading time: 5 minutes
Writing time: 65 minutes

QUESTION AND ANSWER BOOK

Structure of book

Section	*Number of questions*	*Number of questions to be answered*	*Number of marks*
A	6	6	6
B	4	4	42
		Total	48

- Students are permitted to bring into the assessment session: pens, pencils, highlighters, erasers, sharpeners, rulers and one scientific calculator.
- Students are NOT permitted to bring into the examination room: blank sheets of paper and/or white out liquid/tape.

Materials supplied
- Question and answer book of 12 pages, **including answer sheet** for multiple-choice questions.
- A data booklet

Instructions
- Write your **name** and circle your **teacher code** in the space provided above on this page.
- All written responses must be in English.
- Do NOT write on the data booklet supplied.

At the end of the assessment session
- Wait silently for the supervisor to collect your **question and answer booklet**, and your copy of the data sheet provided.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the assessment session room.

Students must not disclose the contents of this outcome; to do so will be a breach of the school VCE guidelines and will be dealt with according to VCAA regulations.

Section A – Multiple Choice Questions

Instructions for Section A

Answer **all** questions in pencil on the answer sheet provided for multiple choice questions.
Choose the response that is **correct** or that **best answers** the question.
A correct answer scores 1, an incorrect answer scores 0.
Marks will **not** be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.

Question 1

Crude Oil is a non-renewable fossil fuel. Once crude oil is extracted from the earth's crust, it is separated into a range of fractions by fractional distillation. Some of these fractions include fuels which are used for most transport vehicles worldwide.

Which of the following fuels will release the greatest amount of energy per gram?

A. octane

B. ethanol

C. propane

D. diesel

Compare heat of combustion values in the data booklet.

Question 2

Australia imports over 90% of the crude oil that it uses to produce fuel for transport. Transport emissions contribute to 19% of total carbon dioxide emissions in Australia. Concerns for emissions, and reliance on other countries for fuel, have sparked interest in looking for renewable alternatives to petrol and petrodiesel. Alternatives include bioethanol, biodiesel, and fuel cells.

Which of the following is a renewable energy resource?

A. Gas obtained from coal seams

B. Ethanol produced from cellulose

C. Hydrogen gas produced by steam reforming

D. Propane produced from natural gas

Question 3

Liquid petroleum gas (LPG) can be used in cars. A major component of LPG is propane (C_3H_8). When 12.00 g of propane is burned in excess oxygen

A. 2220 kJ heat is absorbed from the surroundings.

B. 2220 kJ heat is released to the surroundings.

C. 606 kJ heat is released to the surroundings.

D. 606 kJ heat is absorbed from the surroundings.

Combustion reactions release energy to the surroundings

$50.5\ kJ : 1\ g$

$x\ kJ : 12.00\ g$

$x = 12 \times 50.5\ kJ = \mathbf{606\ kJ}$

Question 4

Which of the following statements is true about an exothermic reaction?

A. The change in enthalpy is positive

B. The temperature of the surroundings will decrease as the reaction proceeds

C. The activation energy of the forward reaction is smaller than the activation energy of the reverse reaction

D. The energy absorbed to break the bonds in the reactants is greater than the energy released during the formation of the bonds in the product

Use the following information to answer Questions 5 and 6

Both ethanol, C_2H_5OH, and methanol, CH_3OH, have been suggested as alternative fuels for transport vehicles. The thermochemical equations describing the complete combustion of ethanol and methanol are given below.

$$C_2H_5OH(g) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) \qquad \Delta H = -1360 \text{ kJ mol}^{-1}$$

$$CH_3OH(g) + \frac{3}{2}O_2(g) \rightarrow CO_2(g) + 2H_2O(l) \qquad \Delta H = -726 \text{ kJ mol}^{-1}$$

Question 5

Separate experiments are conducted in which 1 mol of ethanol and 2 mol of methanol undergo complete combustion. Which of the following statements is incorrect for these experiments?

A. Less water is formed in the combustion of ethanol

B. Both combustion reactions produce the same amount of carbon dioxide

C. The combustion of ethanol consumes more oxygen

D. More heat energy is released from the combustion of methanol

Question 6

What is the total energy released, in kilojoules, when 8.00 L of carbon dioxide is produced by the complete combustion of ethanol at SLC?

A. 220

B. 242

C. 342

D. 439

$$n(CO_2) = \frac{V}{V_M} = \frac{8.00\ L}{24.8\ L\ mol^{-1}} = 0.323\ mol$$

$$2\ mol\ of\ CO_2 : 1360\ kJ\ released$$

$$0.323\ mol\ of\ CO_2 : x\ kJ\ released$$

$$x = \frac{1360\ \times 0.323}{2}\ kJ = \mathbf{220\ kJ}$$

END OF SECTION A

Multiple Choice Answer Sheet

This answer sheet must not be detached from this booklet.

INSTRUCTIONS FOR MULTIPLE CHOICE ANSWER SHEET

Use a **PENCIL** for **ALL** entries. For each question, shade the box which indicates your answer.
Marks will **NOT** be deducted for incorrect answers.
NO MARK will be given if more than **one** answer is completed for any question.
If you make a mistake, **ERASE** the incorrect answer - **DO NOT** cross it out.

1	A	B	**C**	D
2	A	**B**	C	D
3	A	B	**C**	D
4	A	B	**C**	D
5	A	B	**C**	D
6	**A**	B	C	D

Section B – Short Answer Questions

Instructions for Section B

Answer **all** questions in the spaces provided.

To obtain full marks for your responses you should:

- Give simplified answers with an appropriate number of significant figures to all numerical questions; unsimplified answers will not be given full marks.
- Show all working in your answers to numerical questions. No credit will be given for an incorrect answer unless it is accompanied by details of the working.
- make sure chemical equations are balanced and that the formulas for individual substances include an indication of state; for example, $H_2(g)$; NaCl(s)

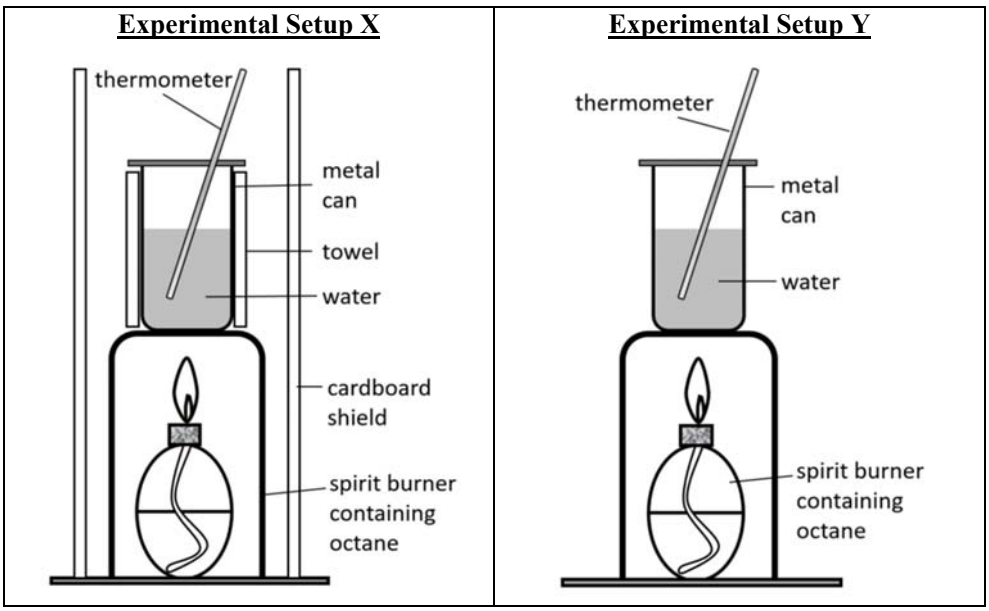
Question 1 (19 marks)

a. Two students, Barry and Fiona, conducted experiments to determine the molar heat of combustion of octane (C_8H_{18}) which is a major component of petrol. Barry and Fiona used the following method:

Method

1. Pour 250 g of water into a metal can. Record the temperature of the water.
2. Fill a spirit burner with octane. Record the mass of the spirit burner and octane.
3. Place the spirit burner under the can of water, light it, and allow it to burn for 5 minutes.
4. After 5 minutes of heating record the temperature of the water and the mass of the spirit burner and octane.

Barry and Fiona sketched diagrams of their experimental setups however they forgot to put their names on the diagrams.



The following table shows the results from Barry's experiment.

Barry's Results	
Temperature of water before heating (°C)	23
Temperature of water after heating (°C)	40
Mass of spirit burner and octane before heating (g)	35.241
Mass of spirit burner and octane after heating (g)	33.941

i. Calculate the amount, in mol, of octane burned in Barry's experiment.

$m = 35.241\ g - 33.941\ g = 1.300\ g$ **(1*M*)**

$n = \frac{m}{M} = \frac{1.300\ g}{114.0\ g/mol} = \mathbf{0.01140\ mol}$ **(1*M*)**

ii. Calculate the heat of combustion, in kJ mol^{-1}, for octane in Barry's experiment.

$\Delta T = 40 - 23 = 17\ °C$ **(1M)**

$q = mc\Delta T$

$= 250\ g \times 4.18\ J\ g^{-1}°C^{-1} \times 17\ °C$

$= 1.78 \times 10^4\ J$ **(1M)**

$\Delta H_c = \frac{q}{n}$

$\Delta H_c = \frac{17.8\ kJ}{0.0144\ mol}$

$= \mathbf{1.6 \times 10^3}\ kJ\ mol^{-1}$

(1M value must be positive)

iii. Fiona used her results to calculate the heat of combustion for octane. Her calculated value was 4000 kJ mol^{-1}. Which of the experimental setup diagrams (X or Y) on page 5 was from Fiona's experiment? Justify your response by referring to the experimental heat of combustion and specific features of the experimental setup.

Set up X **(1M)**

Fiona's ΔH_c is much higher than Barry's therefore Fiona's experiment had less heat lost to the surroundings. **(1M)**

The cardboard shield and towel in setup X would prevent some heat lost to surroundings. **(1M)**

iv. Heat lost to surroundings is a limitation of this experiment that has greatly affected the accuracy of Barry and Fiona's results. Identify one other limitation of the experimental method, as given on page 5. Explain how it could affect the accuracy of the results.

- Not waiting until the max temp is reached before recording. Therefore lower than actual change in temp is recorded decreasing accuracy.
- Water was not stirred, therefore temperature of water may not have been uniform which decreases accuracy of results.
- Position of thermometer was too close to the bottom of the beaker. Therefore a higher than actual change in temp is recorded decreasing accuracy.
- Incomplete combustion resulting in a lower temperature change, decreasing accuracy.
- Experiment not repeated to reduce random errors, decreasing accuracy.
- etc

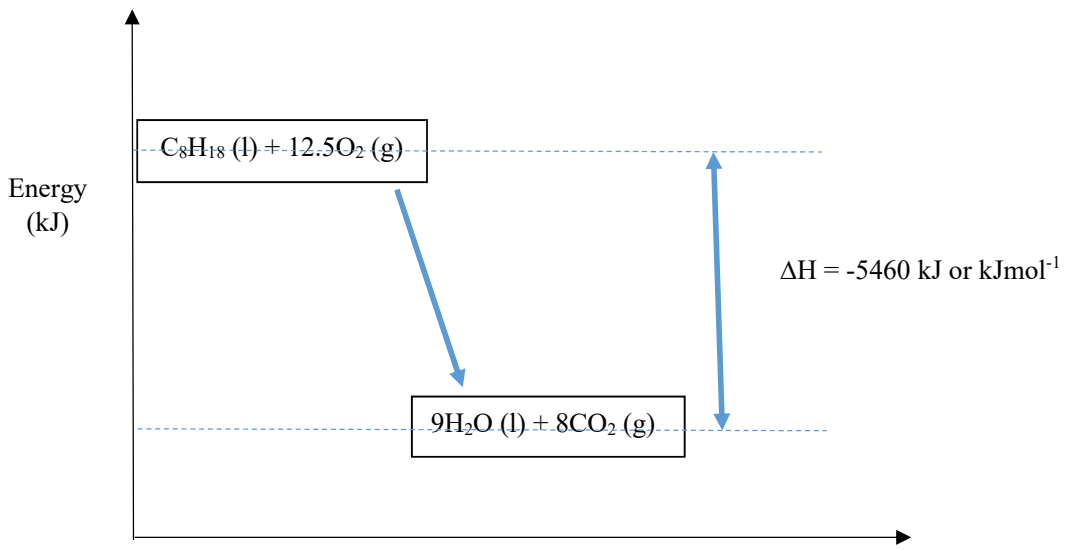
1M for limitation, 1M for <u>explaining</u> how it would affect accuracy.

2 + 3 + 3 + 2 = 10 marks

b. In the cylinders of a car engine, petrol (octane) is ignited, and the energy released is used to drive the car.

i. Write a balanced thermochemical equation for the combustion of liquid octane in excess oxygen.
C_8H_{18} (l) + 12.5O_2 (g) → 9H_2O (l) + 8CO_2 (g); ΔH = -5460 kJ/mol
1M balanced; 1M correct ΔH with correct sign

ii. Draw a diagram on the axis below to show the energy changes that occur during the combustion of octane.



Students can draw an energy profile diagram but are not required to do so for full marks.
1 M – reactants and products labelled
1 M – mole ratio and states included
1 M – ΔH labelled and value included

iii. A car's fuel tank contains 60.0 L of petrol as octane. Calculate the mass, **in kg**, of carbon dioxide produced during the complete combustion of the whole tank of petrol. The density of octane is 0.703 g mL^{-1}.

0.703 kg : 1 L

x kg : 60 L

m(C_8H_{18}) = 60 × 0.703 kg = 42.18 kg **(1M)**

n(C_8H_{18}) = 42.18 × 10^3 g /114 g mol^{-1} = 370 mol **(1M)**

n(CO_2) = 8 × 370 mol = 2960 mol **(1M)**

m(CO_2) = 2960 mol × 44 g/mol = 1.30 × 10^5 g = **130 kg (1M correct answer to 3SF)**

2 + 3 + 4 = 9 marks

Question 2 (7 marks)

Bioethanol (CH_3CH_2OH) is a renewable fuel that is produced from plants such as sugar cane or wheat. Bioethanol can be blended with petrol. E10 is a blend of 10% ethanol in petrol. The following table compares data for each of the components of E10 fuel.

	Octane (petrol)	**Bioethanol**
Mass of CO_2(g) emitted per gram of fuel	Higher	Lower
Mass of CO_2(g) per kJ of energy produced (approx.)	73 000	72 000
Energy content (kJ g^{-1})	47.8	29.7
Energy Density (kJ L^{-1})	33 400	23 400

a) Provide one justification **for** the addition of bioethanol to petrol and one justification **against** the addition of bioethanol to petrol. Use data from the table above to support at least one of your statements.

For:

- Lower carbon emissions /contributing to global warming due to being sourced from plants so some of the CO_2 produced during combustion is absorbed when the plants grow.
- Less CO_2 produced per g/kJ (values from table given) therefore less carbon emission contributing to global warming.

Against:

- Lower energy content/density compared to petrol (values from table given) therefore more fuel needed to travel same distance.
- Large scale production of bioethanol would require large amounts of land that could compete with growing food driving up cost of living.

1M - Use of data
1M -1 valid justification for
1M -1 valid justification against.

3 marks

b) Bioethanol is not carbon neutral. Explain this statement with reference to production, processing and combustion.

- Bioethanol is made from plants. As plants grow CO_2 is absorbed from the atmosphere.(1M)
- When ethanol is burned, the CO_2 is released back into the atmosphere. (1M)
- However, to make bioethanol, the plants must be harvested and the machinery required to do this will produce CO_2 contributing to the overall output of CO_2.(1M)
- Can also mention electricity used to power the factory where bioethanol is processed contributes to the overall output of CO_2.
- Can also mention CO_2 emitted during fermentation.
- Students can show their understanding of first two processes with balanced chemical equations.

3 marks

c) Write a balanced chemical equation for the combustion of ethanol when there is a limited supply of oxygen and carbon monoxide, CO, is the only product containing carbon.

C_2H_6O (l) + $2O_2$ (g) → 2CO (g) + $3H_2O$ (g) 1M balanced with states

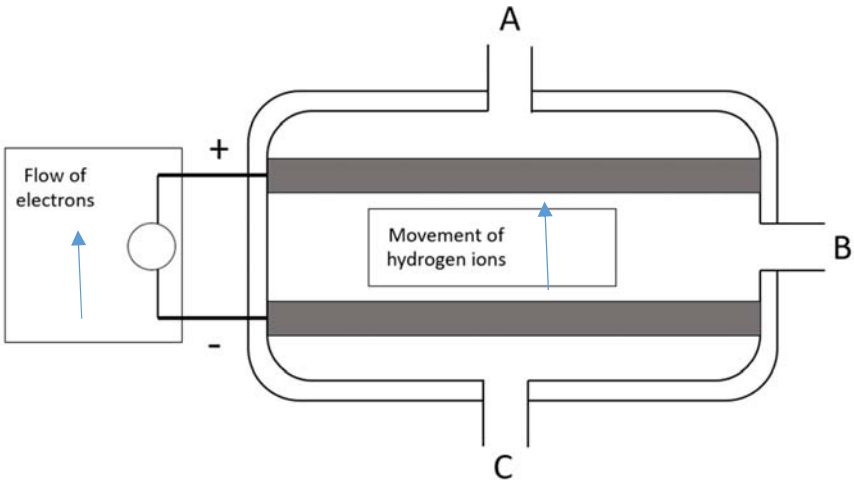
1 mark

Question 3 (8 marks)

Australia's first hydrogen test station opened in 2019 in response to increasing interest in hydrogen-based applications such as the use of fuel-cells in cars.

One such fuel cell, the proton exchange membrane (PEM) fuel cell, consists of platinum-based electrodes and an acidic polymer membrane as its electrolyte. When generating a current, hydrogen fuel is processed at the anode where electrons are separated from hydrogen. At the cathode, electrons combine with oxygen and protons to produce a waste product.

A simplified representation of the PEM fuel cell is shown below:



a. The overall cell reaction is as follows:

$$2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$$

i. At which point, **A**, **B**, or **C**, is hydrogen gas enter the cell?

C

ii. Write an equation for the reaction occurring at the **anode**.

$H_2\,(g) \rightarrow 2H^+\,(aq) + 2e^-$

iii. In the boxes provided on the diagram, use arrows to indicate the movement of hydrogen ions **and** the flow of electrons.

1 + 1 + 2 = 4 marks

b. List two ways that this fuel cell differs from primary galvanic cells.

Any two of:

Constant supply of fuel is required , electrodes are porous, no salt bridge, electrode may contain a catalyst, constant waste H_2O removal

Not contains electrolyte.

2 marks

c. Hydrogen colourless and odourless gas which is extremely flammable.

i. Identify and explain one advantage of using this fuel cell instead of a petrol engine to power a car.

No carbon dioxide emissions produced as a product. H_2O is the only product, therefore does not contribute to carbon dioxide emissions.
More efficient energy conversion in fuel cell so less energy is wasted as heat.
Less noise pollution as fuel cells are much quieter to operate.

Remember that $H_2O_{(g)}$ is also a greenhouse gas.

ii. Outline one safety precaution that would need to be considered when storing hydrogen as a fuel in a car.

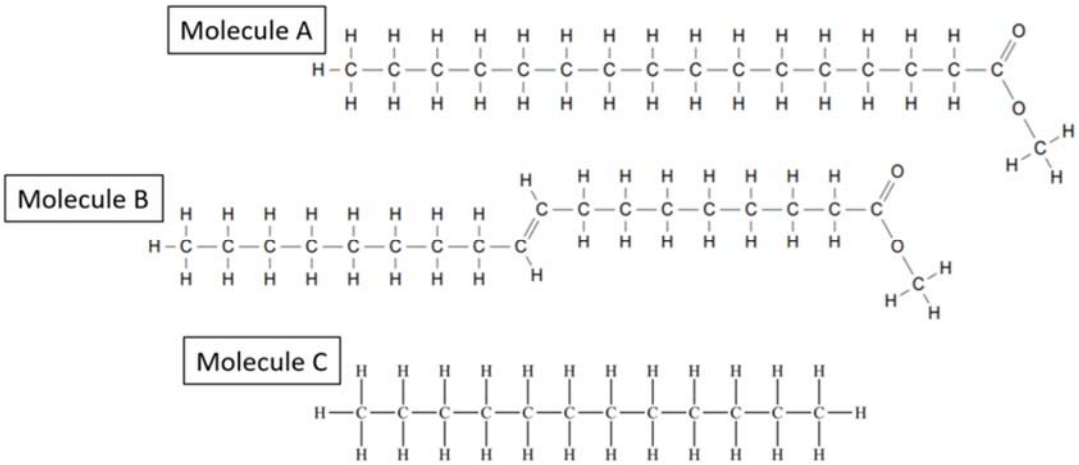
Any one of:
Hydrogen is explosive, so ignition sources must be controlled to a greater extent than petrol.
Hydrogen is stored under pressure requiring an appropriate pressure storage vessel, whilst petrol is not stored under pressure.
Detectors would be required to identify hydrogen leaks., whilst petrol is visible and smells.

1 + 1 = 2 marks

Question 4 (8 marks)

Diesel is a fuel that is often used for heavy-duty equipment and machinery.

a. The images below show the structures of some typical petrodiesel and biodiesel molecules.



i. Complete the following table to identify each of the molecules. The first one has been done for you.

Description	**Molecule**
Straight chain hydrocarbon found in a sample of petrodiesel	*Molecule C*
Unsaturated fatty acid methyl ester (biodiesel molecule derived from plant oils)	B
Saturated fatty acid methyl ester (biodiesel molecule derived from animal fats)	A

ii. Explain why unsaturated biodiesel has a lower melting point than saturated biodiesel.

- Unsaturated biodiesel contain carbon to carbon double bonds that produce kinks in the chain (1M)
- These kinks prevent molecules from packing tightly together resulting in less dispersion forces forming between molecules. (1M)

1 + 2 = 3 marks

b. A winter service vehicle is a vehicle that is specially designed to clear snow from roads in areas that experience heavy snowfall during the winter months.

The table below shows the difference in some properties between petrodiesel and biodiesel.

Fuel	**Cloud Point**	**Hygroscopic nature**	**Viscosity**
Petrodiesel	Lower	Less	Lower
Biodiesel	Higher	More	Higher

iii. Use the information in the table above to explain why winter service vehicles use petrodiesel and **not** biodiesel.

Petrol diesel has a lower viscosity therefore will flow more easily along fuel lines.
Viscosity of fuels increase as temperature decrease so it is better to choose a less viscous fuel during the colder months as fuels need to flow easily along fuel lines for better performance.

Cloud point measure the temperature at which solid crystals of fuel start to form. Since petrol diesel has a lower cloud point, then it is less likely to solidify in colder climates.

iv. Select one of the properties, cloud point, hygroscopic nature, **or** viscosity. Explain the difference in this property between petrodiesel and biodiesel in terms of structure and bonding.

Property:

Explanation: polar group in biodiesel (1M), dipole-dipole, and dispersion forces in biodiesel and dispersion forces only in petrol diesel (1M), link to how intermolecular forces affect property ie. compare fuels properties (1M)

Hygroscopic should reference hydrogen bonding between biodiesel and water.

2 + 3 = 5 marks

END OF QUESTION AND ANSWER BOOKLET